

Array Method

Array Methods

- **forEach**
- **map**
- **filter**
- **some**
- **every**
- **reduce**

forEach

arr.forEach(some function definition or name);

forEach

```
let arr = [1, 2, 3, 4, 5];
```

```
function print(el) {  
  console.log(el);  
}
```

```
arr.forEach(print);
```

```
// OR
```

```
arr.forEach(function(el) {  
  console.log(el);  
});
```

Map

let newArr = arr.map(some function definition or name);

```
let num = [1, 2, 3, 4];  
  
let double = num.map(function(el) {  
  return el*2;  
});
```

Filter

let newArr = arr.filter(some function definition or name);

```
let nums = [2, 4, 1, 5, 6, 2, 7, 8, 9];  
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let even = nums.filter( (num) => (num % 2 == 0) );
```

Every

Returns true if **every element of array gives true** for some function. Else returns false.

arr.every(some function definition or name);

```
[1, 2, 3, 4].every( (el) => (el%2 == 0));  
false  
[2, 4].every( (el) => (el%2 == 0));  
true
```

Some

Returns true if **some elements of array give true** for some function. Else returns false.

arr.some(some function definition or name);

```
[1, 2, 3, 4].some( (el) => (el%2 == 0));  
true  
[1, 3].some( (el) => (el%2 == 0));  
false
```

Reduce

Finding **Maximum** in an array

```
let nums = [2, 3, 4, 5, 3, 4, 7, 8, 1, 2];

let result = nums.reduce( (max, el) => {
  if(el > max) {
    return el;
  } else {
    return max;
  }
});
```

Default Parameters

Giving a default value to the arguments

```
function func (a, b = 2) {
  //do something
}
```

```
function sum(a, b = 3) {
  return a + b;
}
```

```
sum(2); //5
```

Spread

with Array Literals

```
> let arr = [1, 2, 3, 4, 5];  
< undefined
```

```
> let newArr = [...arr];  
< undefined
```

```
> newArr  
< ► (5) [1, 2, 3, 4, 5]
```

```
> let chars = [..."hello"];  
< undefined
```

```
> chars  
< ► (5) ['h', 'e', 'l', 'l', 'o']
```

Spread

with Object Literals

```
let data = {  
  email: "ironman@gmail.com",  
  password: "abcd",  
};  
  
let dataCopy = { ...data, id: 123 };
```

Rest

Allows a function to take an indefinite number of arguments and bundle them in an array

```
function sum(...args) {  
  return args.reduce((add, el) => add + el);  
}
```

Destructuring

Storing values of array into multiple variables

array

```
let names = ["tony", "bruce", "steve", "peter"];  
let [winner, runnerup] = names;  
console.log(winner, runnerup); // "tony" "bruce"
```

Destructuring

Objects

```
const student = {  
  name: "karan",  
  class: 9,  
  age: 14,  
  subjects: ["hindi", "english", "math", "science", "social studies"],  
  username: "karan123",  
  password: 1234,  
};  
  
const { username: user, password: pass } = student;  
  
console.log(user); // "karan123"
```